Innovative Winners: Variation of Technological Innovation & Fitness in Industrial Policy

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Abstract

This paper investigates the role that governments may play as strategic signalers altering the variation of products in the marketplace. More specifically, I argue that markets can be persuaded by signals from the government. I propose that market dominance may not be solely determined by intrinsic product qualities, like innovative-ness, but that governments can strategically signal to markets about product fitness, favoring goods that meet operational requirements independently of their inherent technical merit. To flesh out these mechanisms, I develop a Bayesian Persuasion model in which governments signal to markets on two dimensions: product fitness and innovative-ness. Equilibrium analysis of the model reveals that while government signaling about which products it thinks will succeed is effective, that when governments try to make predictions about intrinsic product innovative-ness, there are suboptimal product-level outcomes. These findings suggest that the effectiveness of industrial policy depends not on whether governments intervene, but on whether and how they restrict their signals to what they can credibly observe.

1 Introduction

What explains the variation in technological innovation in the business-to-government (B2G) market? Often referred to as the procurement or contract market, the B2G market is one of the most lucrative markets both within the United States and abroad. Governments often have an incentive to encourage innovation in the B2G market to support national security (Mowery, 2009). For instance, the Ukrainian government, in its defense against Russia, has poured countless resources into defense innovation, allowing for some to say that the nation has "redefined modern warfare," (Nabukhotny, 2025). Other countries, like the United States have consistently struggled with defense innovation, irrespective of a large resource pool dedicated to technological innovation (Steinbock, 2014).

Corporations and governments can often be aligned in their goals to create more innovative products. Yet divergent outcomes exist when governments pursue industrial policies directed at spurring innovation. In fact, what I will argue in this paper, is that by restricting what governments can signal to just what they can observe, namely product fitness, produces better innovative outcomes than broader intervention. This is complicated by a problem of selection bias. Assuming, in the B2G market that the government contracts the best, most innovative and successful, products, then observing the ones that are not successful (i.e., product failures) becomes nearly impossible. This requires developing a careful definition of innovation in order to hypothesize about not just the products that are observable but also the ones that do not.

What I propose is that an innovation may be defined as something that improves upon a previous product but whose success in the market is a function of the innovation's "fitness". The fitness of an innovation depends on the market environment, which in turn is influenced by innovations that are already in the market and government reactions to the rise and fall of those innovations. The way that governments react to innovations can be thought of as shaping them to enhance the competitiveness of a certain product, sometimes independently of their intrinsic innovative technological quality.

Specifically, I propose that market dominance is not determined solely by product innovative-ness, because governments can strategically alter market behavior to favor certain goods. The means by which I propose governments can alter market behavior is through what I describe as a product's "fitness." Drawing on evolutionary biology, fitness is used to describe how adaptable a product is to the market (Hochberg et al., 2017). I argue, that by adjusting signals to the market about a product's fitness, governments can play an integral role in the success of an innovation. And perhaps more importantly, I argue that government signaling about product fitness will lead to greater instances lack-luster product innovation. In order to flesh out the theoretical aspects of my argument, I formulate a formal model of the B2G market in which governments can signal information about product fitness and innovative-ness, demonstrating that the degree of governments signals determines innovative outcomes. What I find is that government signaling about product fitness increases the likelihood that high-fitness products succeed, but when governments add signals about innovative-ness, the market outcomes falls short of the social optimum.

The paper proceeds as follows: First, I outline the existing literature on innovation and public-private partnerships. I then turn to discussing the distinction between innovative-ness and fitness and then proceed to introduce the formal model. Section 3 goes into the model setup and Section 4 discusses the equilibrium analysis and propositions. Section 5 concludes the paper by discussing the implications of the formal analysis and offering avenues for additional research on the topic. Note that throughout the paper I will refer to government intervention and public-private partnerships interchangeably. Both are meant to describe the process of industrial policy directed at spurring innovation.

2 Industrial Policy and Government Signaling

Governments seek economic growth not only for electoral reasons but also for domestic and international welfare (Tourinho, 2021). While authors debate the extent to which innovation spurs growth (Schumpeter, 1942; Duesenberry, 1956), the consensus seems to be that technological progress has led to development and increased GDP (Scherer, 1986). However, firms have incentives not to innovative. As Arrow (1993) describes, innovations are uncertain by nature–if everything about an innovation were known, it would not be an innovation. It follows then that the supply of innovations always outweighs the market demand because their economic potential is known to the innovator but not to the market (Arrow, 1993). Innovations are thus incredibly risky, and many firms who take such risks fail (Eisenmann, 2021).

Additionally, proven products tend to persist even with minor flaws because replacements are costly and do not necessarily produce verifiably better products. Akcigit and Goldschlag (2023) find that while inventors are more concentrated in old, large, incumbent firms, those at younger firms tend to produce more impactful innovations. This compliments Arrow (1993), who finds that large firms are often biased against originality because of a greater information loss between inventors (R&D departments, etc.) and the financiers (higher-ups in the firm); a gap that is smaller in younger firms where the two roles are often the same individuals (Arrow, 1993).

This information transfer problem is central to this paper. Arrow (1993) highlights that R&D is affected by both the development profitability function of the product and the availability of capital—each conditioned on information transfer within and external to firms. Small firms may better estimate the development profitability function but struggle to communicate product value to external capital sources, while large firms have internal capital but struggle to justify funding innovations internally.

This paper argues that governments can ease the information transfer problem both solving some of the disincentives for firms to innovate. Specifically, I argue that governments can act as predictors of product success; thereby inducing success by signaling their assessments to the market. The basis for this kind of government information signaling does not come from superior knowledge about a product's inherent technical qualities, if anything firms would retain that advantage. Instead, I argue it comes from the government's accumulated knowledge of operational requirements, procurement history, and product deployment needs.

In the national security context particularly, governments may intentionally obfuscate or hide the use-cases for products from markets in order to maintain classified material. This asymmetry, I argue, justifies the government's role as a sender in a persuasion framework: the government holds private information about product fitness and can credible signal that to agencies evaluating their procurement decisions. In this case, the agency can be thought of as the agent in a principal-agent framework where the government is the principal. Thus, the agency can follow the directives of the government but have incentives to prioritize their own goals and the government, knowing this may have incentives to obscure information Jensen and Meckling (1976).

Nonetheless, critiques about government intervention in the market exist. Howell (2024) argues that "government has a key role to play in the type of failure-tolerant, open-ended research that yields breakthrough inventions," while there is "less evidence that subsidizing financial intermediaries is useful." Common concerns in the literature include governments "locking-in" insufficient outcomes or picking insufficient winners (Lerner, 2012), crowding out private competition through procurement (Howell et al., 2024). There is also some research that indicates that government tax incentives may not have an effect on innovation (Goolsbee, 1998; Van Reenen, 2021). These failures largely stem from a misalignment of policy and information. Akcigit et al. (2022) argue that the instrument through which government intervention occurs should be aligned with what the government can observe. Howell (2024) further notes that government interests (i.e, health, natural security) may not align with market incentives, potentially decoupling product success from inherent quality.

These critiques all suggest that government intervention largely fails when instruments of the intervention are misaligned with the government's informational capacity. It is my argument that signaling addresses this directly. Other types of intervention such as subsidies or procurement guarantees require the government of act on their assessment of both fitness and innovative-ness, while the signaling mechanism I put forward, can be restricted to what governments can observe; namely whether they think a product will succeed/fit operational requirements. This distinction bears out in the model's equilibrium. That is, when the government only signals on fitness, outcomes improve moderately. But when innovative-ness is incorporated into the signal, the equilibrium falls short of the social optimum, precisely because governments lack the information to make credible claims about inherent product quality.

2.1 Innovation and Bayesian Persuasion

The idea that an innovation's success may not be determined solely by its intrinsic qualities is not new. Schumpeter (1942) argued that "Innovation is the market introduction of a technical or organisational novelty, not just its invention". In other words, success depends on the market displacement of older goods, not just the invention alone. The smartphone camera is a clear example: built from existing components, it succeeded through market fit rather than pure innovative-ness (Ratner, 2019).

Fitness, borrowed from evolutionary biology, describes a mutation's relative ability to survive in its environment (Hochberg et al., 2017). Following Schumpeter's own formulation that innovations are "industrial mutations" (Schumpeter, 1942), I argue that products similarly have fitness, or a relative ability to succeed in the market independent of their intrinsic qualities. The adoption of the 5.56×45mm ammunition by NATO during the Cold War (Zhou, 2016) illustrates this: the ammunition persists as the global military standard not solely due to ballistic superiority (Courtney et al., 2017), but because supply chains and interoperability make it highly fit in its procurement environment.

Fitness and innovative-ness are conceptually and informationally distinct. Fitness is relative, it describes how well a product meets the requirements of a particular environment (or market in this case). To some extent, it is observable by the government, either through operational requirements or procurement history. To that degree, it is not purely intrinsic to the product, but it depends on the government's needs. Innovative-ness on the other hand is intrinsic to the product. It describes the degree to which a product constitutes a genuine technical advance (or recombination in Schumpeterian terms) and information about innovative-ness is largely privately held by the firm or the market but over long periods of time. Governments may try to observe innovative-ness with proxies like patents or investment in R&D, but the firm largely retains the informational advantage.

This informational asymmetry is what motivates the model's structure. The government signals about fitness, while innovative-ness enters through the expected correlation between the two dimensions. This is what motivates restricting the government's signal to just fitness in the model. Table 1 captures the key distinctions.

Fitness	*Innovative-ness*	
	High	**Low**
High	unicorns	lack-luster products
Low	risky products	failures

Table 1: Product Innovative-ness vs. Fitness

My model draws from Kamenica and Gentzkow (2011)'s Bayesian Persuasion, where a sender commits a signal that shapes a receiver's actions through the receiver's beliefs. In my model, the government is the sender and the agency the receiver. Laffont and Tirole (1993) apply principal-agent theory to model the relationship between a regulator and firm under incomplete information to illustrate the point that contracting becomes easier when the firm knows its costs and capabilities more than the regulator does. Their key finding is that when the firm has private information, it is still possible for the government to create contracts that convince the firm to act in the public's best interest. My model takes a complementary approach, where governments are the ones with private information and they choose to strategically reveal information about product quality in order to convince agencies to contract with those product's firms.

Likewise, Hoppe and Schmitz (2013) model the provision of a public good across two scenarios, public-private partnerships (PPPs) and traditional procurement, in order to test whether the government's choice of provision distorts incentives for firms to innovative. They find that the government's preferred choice depends on information-gathering costs, costs of innovation, and the degree to which effort is contractable (Hoppe and Schmitz, 2013). Specifically, for PPPs in which the government shares both building and design of the product with a firm, they find that PPPs provide stronger incentives for firms to invest in innovation, but not necessarily in quality innovations. My model asks a very similar question to Hoppe and Schmitz (2013), except I make the concept that information transfer from a government may effect product-level outcomes explicit using a signal-type model instead one of public-good provision.

Finally, Bergemann and Morris (2016) consider the problem of an "information designer" in the context of a Bayesian persuasion model where are multiple ($n > 2$) players by introducing Bayesian correlated equilibria. Their finding is that with more players, if the sender gives the players more information about the state of the world, the sender actually constrains the set of outcomes it can implement. My model addresses a related design problem where with just one receiver extra signal information distorts the potential outcomes, but a mutli-receiver extension is a logical next step for the model. Some like Little (2023) raise valid concerns about Bayesian persuasion models with information designers as they require sender to pre-commit to sending certain signals. Little (2023) argues that this may not map onto the real world in a more useful way than classic perfect Bayesian equilibrium (PBE) designs. This is a fair critique, yet the most obvious example of when a sender might want to pre-commit is in the case of regulation and policy. Particularly, in the context of this model, where governments want an outcome from their agencies that aligns with their signal design, it seems fitting to allow them to fully reveal their signal design in a Bayesian persuasion context. To better illustrate this point, I also introduce an alternative model design in the next section where firm entry is a function of government signaling. The analysis of the alternative model reveals that pre-committing to signals is necessary to induce firm entry which ultimately results in product innovation.

3 The Model

Suppose there is a firm (F) that creates a product which has both a fitness, θ, and an innovative-ness, τ. Suppose there is also a government, G, who commits to sending a signal to an agency, A, about a product, i. The agency's beliefs and actions are influenced by the information transmission from G. There are two states of the world, representing the inherent fitness of the product: $\theta_i \in \{H, L\}$ where H represents high fitness and L indicates low fitness. At the start of the game, G observes θ but the agency does not. Instead, A has some prior belief that $Pr(\theta = H) = \pi$ and $Pr(\theta = L) = 1 - \pi$. The government commits to sending a signal, σ that maps θ onto m, where $m \in \{g, b\}$ and g is a positive signal (the product is high fitness) and b is a negative one (the product is low fitness). Let $Pr(m = g|\theta = H) = \alpha$ and the $Pr(m = b|\theta = L) = \beta$, where $\alpha \geq \beta$, such that more fit products receive positive signals. Let $\alpha, \beta \in [0, 1]$ such that if $\alpha = \beta = 1$, then signal is fully informative. The model takes inspiration from Kamenica and Gentzkow (2011) Bayesian Persuasion model and equilibrium design. Kamenica and Gentzkow (2011) key insight is that at times self-interested parties can design what information to reveal to the receiver and benefit, even when the receiver knows it may be being manipulated.

After observing the signal, the agency updates its beliefs using Bayes Rule, such that it's posterior beliefs are:

$$\mu_g = \frac{\alpha\pi}{\alpha\pi + (1 - \beta)(1 - \pi)} \tag{1}$$

$$\mu_b = \frac{(1 - \alpha)\pi}{(1 - \alpha)\pi + \beta(1 - \pi)} \tag{2}$$

Note that $\mu_g \geq \pi \geq \mu_b$ whenever $\alpha \geq 1 - \beta$. Thus the signal is Bayes Plausible; it perseveres $Pr(m = g)\mu_g + Pr(m = b)\mu_b = \pi$.

A then decides whether or not to procure the product, $a \in \{\text{procure}, \neg\text{procure}\}$. The agency's payoff is thus:

$$u_A = \begin{cases} \theta - c & \text{if } a = \text{procure} \\ 0 & \text{if } a = \neg\text{procure} \end{cases}$$

where $c \in (0,1)$ is the cost of procuring the product (this can also be thought of as the contract value). Here, I choose to reduce the scale of the contract value to within the bounds of $(0,1)$ but truthfully the contract value could be much greater—but perhaps not infinity large since agencies have budgets that cap their spending. Thus the agency's decision rule for whether to procure a product can be formulated as: $E[\theta|m] \geq c$ or when $\mu_m\theta_H + (1-\mu_m)\theta_L \geq c$. The government prefers the high fitness products to be procured, thus their payoff can be formulated as:

$$u_G = \begin{cases} 1 & \text{if } a = \text{procure and } \theta = H \\ 0 & \text{if otherwise} \end{cases}$$

G chooses α, β to maximize EU_G subject to the agency's decision rule. Note that G does not care about accidentally signaling about low-fitness types, but only about not missing high-fitness ones. The model thus far does not specify the agency's preferences over product fitness, but it makes little sense to think that the agency would prefer to procure unsuccessful products, so I assume that $\theta_H > \theta_L$, which allows me to simplify the agency's procurement rule by setting $\theta_H = 1, \theta_L = 0$. Thus, I define the agency's procurement rule, $\mu_m > c$ as a function of two regions in posterior belief space: 1) Procure space: $\mu_m \geq c$, 2) Reject space: $\mu_m < c$.[1]

[1] As an aside, when $\theta_L > 0$, the agency's procurement rule does change but it is now easier to convince, since it has less to lose from procuring a low fitness type. Yet, if the goal is to test the case where government signaling about high fitness types works, I argue that going after the cases where the agency is more difficult to persuade is more meaningful.

4 Equilibrium Analysis

4.1 Baseline Model

An equilibrium in this game consists of optimal signals for the government to send, (α^*, β^*), the agency's optimal action for each signal, $a^*(m)$ where $m \in \{g, b\}$, and the agency's posterior beliefs, μ_g, μ_b. For an equilibrium it must be the case that $a^*(m)$ maximizes u_A given μ_m, α^*, β^* which maximizes $E[u_G]$ given $a^*(\cdot)$, and beliefs are consistent with the Bayes plausibility constraint whenever a signal is sent with positive probability. Given this formulation, there are two possible scenarios, either $\pi \geq c$, meaning that the agency would procure regardless of the signal sent or $\pi < c$, implying that the agency would not procure under their current priors. Consider then Lemma 1:

Lemma 1. *If $\pi \geq c$ then the agency procures regardless of signal, G doesn't design a signal, yielding $E[u_G] = \pi$. If $\pi < c$ the agency only procures if G's signal shifts A's posterior above c for at least the positive message, $m = g$.*

Given A's payoff function when $\pi > c$, then the agency procures regardless of if or which signal is sent. G, knowing this, has no incentive to design a signal. Regardless of what G chooses to signal, it gets $EU_G = \pi$ or the prior belief of A's that the product is of high fitness (since A's decision to procure is solely based on it's prior, $\pi > c$). On the other hand, when $\pi < c$, the agency has no incentive to procure under their prior and G, therefore, must design a signal that will persuade or shift A's posterior belief about the product above c for at least $m = g$. G's optimal signal is design by finding the α^* that pushes $\mu_g \geq c$ and the β^* that $\max(\text{prob}(m_g|H))$. Given Lemma 1 consider then the following proposition where $\pi < c$:

Proposition 1. *There exists a unique PBE when $\pi < c$, the optimal signal is $(\alpha^*, \beta^*) = (1, 1 - \frac{\pi(1-c)}{c(1-\pi)})$ which shifts beliefs such that $\mu_g = c$.*

When $\pi < c$, G must signal optimally to push $\mu_g \geq c$. If G pushes $\mu_g = c$ and $\mu_b = c$

then it gets in equilibrium, $\alpha^*\pi = \pi$ and $(1-\alpha^*)+\beta(1-\pi)$, respectively. If G pushes $\mu_g < c$, then the agency rejects yielding nothing for G. If G pushes $\mu_g > c$, then the agency still procures on $\theta = H$ like in the case when $\mu_g = c$. The only case when increasing μ_g above c would help G is when it also increased μ_b above c, causing the agency to procure when the signal is negative too. However, by the Bayes Plausibility constraint any increase in μ_g causes a decrease in μ_b, therefore G gains nothing from $\mu_g > c$ and in equilibrium sends m^* to shift $\mu_g = c$.

Now consider the optimal α^*, β^* that can achieve m^*. In equilibrium, G's objective function yields $\alpha\pi$, which is increasing in both α and π. Therefore, G wants to set α as great as possible with the constraint of $\alpha \in [0, 1]$ and to satisfy $\mu_g = c$. Thus when $\alpha = 1$ then for $\mu_g = c$ it must be the case that $\beta = 1 - \frac{\pi(1-c)}{(1-\pi)}$ for $\beta \in (0, 1)$. Given α^*, β^*, the agency is indifferent between procuring when the signal is $m = g$, but I assume that procuring is a weakly preferred to not. When $m = b$, since $\alpha^* = 1 > \beta^*$ and $\mu_b < c$ the agency strictly prefers not to procure after a negative signal, $m = b$. Beliefs are consistent with Bayes Rule and all possible messages are sent with positive probability, therefore they are no off-equilibrium path beliefs that must be specified. All proofs are provided in the Appendix.

The key result from this equilibrium, as it relates to the hypotheses put forward previously, is twofold. First, when $\alpha^* = 1$, trivially, all high fitness products are procured. Second, and more interesting, is that when this is true, is that some low fitness products are also procured. The probability the agency procures given the type is low, given the signal $m = g$, is $1 - \beta^* = \frac{\pi(1-c)}{c(1-\pi)}$ which is greater than 0, implying that when G designs a signal, the agency sometimes procures even the low fitness products, even though the government strictly prefers if only high fitness ones get procured. This is true whenever $\pi, c > 0$ which I assumed is the case.

Note that this baseline model does not incorporate any realization about product innovativeness. The baseline model, in this case, can be thought of as a restricted view of the world, where governments only care about the products that they think will be successful (i.e., fitness). The agency still makes procurement decisions knowing that the government may have an incentive to manipulate signals about product type, resulting in the procurement of sometimes suboptimal products. But, ideally when governments are pursing industrial policies directed at innovation, they also care about whether the highly innovative products get procured or not. The next step in the model then is to introduce innovative-ness as another component of the signal that G can send to A. First however, it may be necessary to develop intuition as to why the government must pre-commit to signals as they do in the baseline model.

4.2 Alternative Perfect Bayesian Equilibrium

If the government only cares about high fitness types getting procured then it might be natural to ask, why try and persuade the agency at all? In other words, the Bayesian persuasion model presented thus far allows the government to send messages about both high and low fitness products (since β in equilibrium is greater than zero). It could be argued that if the government only cares about the high fitness types getting procured that the government should just fully reveal types by signaling either just α or $\alpha = 1, \beta = 0$. In fact, this would revert the game to a simpler Perfect Bayesian Equilibrium (PBE). To illustrate why Bayesian Persuasion is necessary, I construct a separate formal model where firms are strategic and can choose to enter the market as a function of expected procurement outcomes.

The model assumes that firms choose to enter or stay out of the market, prior to learning the type of product, where entering requires paying some cost, k_H for high fitness products and k_L for low fitness products (assume like in the previous model $k_H > k_L$). I solve the game for a PBE, where the government chooses either a positive or negative signal given the type and entry, and also for Bayesian Persuasion, where the government pre-commits to a signal containing information about both types. This extension is relatively simple: all players remain the same except that firms now start the game and in the PBE case, governments signal based on learning of the type of product not pre-committing. I provide a game tree for this alternative model in the Appendix along with all proofs for both PBEs and Bayesian Persuasion.

Proposition 2. *There exists a PBE where firms enter, G sends $m = g$ for $\theta = H$ and $m = b$ for $\theta = L$, and the agency procures only when $m = g$. If A observes a product entered they update their beliefs based on Bayes rule and if they do not enter they believe product is $\theta = L$. It is also the case that in this PBE, firm entry condition is harder to meet than in the Bayesian Persuasion case.*

Upon observing the signal $m = g$, A updates their beliefs that $Pr(\theta = H|m = g) = 1$ and upon observing $m = b$ that the $Pr(\theta = L|m = b) = 1$. Given these beliefs, the agency finds it optimal to procure only on the positive message so long as $\theta_H > c > \theta_L$. G finds it optimal to signal $m = g$ only on the $\theta = H$ type since it cares only about high fitness products being procured, so any such deviation would be not beneficial. Finally, F's decision to enter is a lottery. Entrance is optimal whenever $\pi c \geq \pi k_H + (1 - \pi)k_L$ or whenever the expected value of producing a high fitness product outweighs the risk of being a low fitness product.

There are other PBE that exist for this alternative model but they are all unstable to intuitive criteria constraints. Pooling on entry where the government sends only positive messages is a PBE but violates intuitive criteria because G would always want to send a negative message about low type products instead. Similarly, pooling on staying out where the agency never procures regardless of m requires off-path beliefs to be pessimistic—that is A must believe that only the firms with high type products would enter and they'd still have to not procure those products. Lastly, mixing can occur where firms enter probabilistic and G sends $m = g$ to $\theta = H$ and vice versa for low types. However, in equilibrium this requires firms to be strictly indifferent over the expected value of high fitness products and low fitness products. Yet, this is the exact situation that Arrow (1993) describes where firms deciding between creating new innovations and staying with 'good enough' products tend to side with latter. So in the PBE case, entering the market is a gamble for firms, and results in fewer firms entering the market with new products as c decreases, or the expected contract value, and as the cost of high product type production, k_H increase.

Consider then the Bayesian persuasion version of this alternative model. Firms choose to enter before knowing their type, like before, so now their payoff is a function of the likelihood of their product being a certain type and the probability the agency procures given G's pre-committed signal containing the probabilities α, β. Thus I derive the firm's entry condition as:

$$c(\alpha\pi(1-\pi)(1-\beta)) \geq \pi k_H + (1-\pi)k_L \tag{3}$$

Essentially, F has to weigh the expected contract revenue when their product is procured against the expected production costs.

In order to induce firm entry and agency procurement, G now must solve two separate conditions: 1) set $\mu_g > c$ and 2) set $c(\alpha\pi(1-\pi)(1-\beta)) \geq \pi k_H + (1-\pi)k_L$ while maximizing α, β subject to $\alpha, \beta \in [0,1]$. From the baseline restricted model (Proposition 1), I showed that condition 1 is met when $\alpha^* = 1, \beta^* = 1 - \frac{\pi(1-c)}{c(1-\pi)}$. Plugging in these equilibrium values into the firm's entry condition, Condition 2 can be met as well:

$$c(\pi + (1-\pi)(1-(1-\frac{\pi(1-c)}{c(1-\pi)}) \geq \pi k_H + (1-\pi)k_L \tag{4}$$

$$\Rightarrow \pi \geq \pi k_H + (1-\pi)k_L \tag{5}$$

Thus in equilibrium, G can send signals with the exact same probability as in the baseline model and induce procurement and firm entry under the condition in Equation 5. Consider this in comparison to the condition for firm entry in the PBE of this alternative model, $\pi c \geq \pi k_H + (1-\pi)k_L$. Since right hand side of both inequalities are the same and since $c \in (0,1)$, then $\pi c < \pi$ and so the entry condition in the Bayesian Persuasion case is easier to satisfy than in the PBE. What this implies is that pre-commitment of signals by G sustains more firm entry than in a game where G only signals about the types it wants. Firm entry is important because it means more products get produced for G to observe. Ultimately too if the purpose of industrial policy like described in the empirical motivation for this model, then without firm entry, innovation is surely not guaranteed. Encouraging firm entry is a necessary step to product innovation. I now turn to adding in innovation into the model.

4.3 Unrestricted Model - Adding Innovative-ness

From the equilibrium analysis of the original model, I know that even when governments signal optimally, agencies may still procure low fitness products. Now, I turn to allowing the government to care about product innovative-ness as well as fitness, even when they cannot know inherent innovative-ness. Consider the model when the state space is expanded to include another dimension, τ (representing the true innovative-ness of a product i, such that any state of the world, $\omega = (\theta, \tau) \in \{H, L\} \times \{H, L\}$). G observes θ, as in the original model, but not τ. A observes neither θ nor τ. Yet τ is correlated with θ and G makes a prediction about how the two parameters are correlated and thus can signal about innovative-ness without observing its true value. I define this correlation as:

$$Pr(\tau = H|\theta = H) = \rho \tag{6}$$

$$Pr(\tau = H|\theta = L) = \delta \tag{7}$$

where $\rho > \delta$, both $\in (0, 1)$. Note here, that assuming $\rho > \delta$ is to assume that there is a positive correlation between θ and τ conditionally. Thus, A's marginal prior on i remains the same, $Pr(\theta = H) = \pi$, but now the joint prior are shown in Table 2.

The agency's beliefs about θ after observing m are the same as in the baseline model, but now take into account τ. Given the law of total probability, it must be the case that the probability the product is high innovative-ness given a positive signal must equal the sum of

Fitness	***Innovative-ness***	
	$\tau = H$	$\tau = L$
$\theta = H$	$\rho\pi$	$(1-\rho)\pi$
$\theta = L$	$(1-\pi)\delta$	$(1-\pi)(1-\delta)$

Table 2: A's Joint Priors about Product i

the joint probability that the product is high innovative-ness and fitness and the probability that the product is high innovative-ness and low fitness given a positive signal. Therefore the posterior belief of A after observing $m = g$ is $\mu_g(\rho - \delta) + \delta$. Let this posterior belief that $\tau = H$ be v_g. Similarly then let v_b be the posterior belief that $\tau = L$ and equal $\delta + (\rho - \delta)\mu_b$. Since $\rho > \delta$ and $\mu_g > \mu_b$, then $v_g > v_b$. Generically let $v_m = \delta + (\rho - \delta)\mu_m$ be the posterior beliefs of A after any signal is set.

Payoffs are similar to that in the baseline model, except players now weight their payoffs by how much they value innovation. Let $A's$ weight on innovation be $\lambda > 0$ and let $G's$ weight be $\Gamma > 0$. Therefore, after observing m, A procures if and only if $\mu_m + \lambda(v_m) \geq c$ or when the sum of their posterior beliefs about the product are greater than the contract value. Substituting in for v_m, the agency's procurement rule is $\mu_m \geq \frac{c-\delta\lambda}{1+\lambda(\rho-\delta)}$. I define the threshold for procurement as $\hat{c} = \frac{c-\delta\lambda}{1+\lambda(\rho-\delta)}$. Like before, the government's payoff is now a factor of the agency's procurement rule in addition to it's weight on innovation, Γ. Therefore consider the following proposition:

Proposition 3. *There exists a PBE in which $\rho > \delta$, $\pi < \hat{c}$, and G sends a signal m^* to persuade A to procure when $\alpha^* = 1, \beta^* = 1 - \frac{\pi(1-\hat{c})}{\hat{c}(1-\pi)}$.*

The proof of this proposition follows the same logic of the baseline model, replacing c with $\hat{c}$. A won't procure when their priors are such that $\pi < \hat{c}$, so G will design a signal that moves $\mu_g = \hat{c}$ and set optimal α, β accordingly. Since G cares about high fitness products getting procured still, α is set maximally at 1 for a fully informative signal, and β^* is found by plugging in $\alpha^* = 1$ and solving, which gives us $\beta^* = 1 - \frac{\pi(1-\hat{c})}{\hat{c}(1-\pi)}$. Note that since $\hat{c} < c$ that $\beta^*_{baseline} > \beta^*_{unrestricted}$, which implies that adding more information to the signal makes the agency easier to persuade.

The expected innovative-ness of products procured is now $v_g = \delta + (\rho - \delta)\hat{c}$, since only products with a positive signal get procured, as in the baseline model. Note that in the baseline model, innovative-ness was never signaled on. Therefore the expected innovative-ness of procured products was simply the weighted average of the agency's priors, $\delta + (\rho - \delta)\pi$ or when G was signaling optimally in the baseline and $\mu_g = c$, thus the expected value of innovative-ness for a procured product is $\delta + (\rho - \delta)c$. Consider now the difference, Δ, in expected value of innovative-ness between the baseline and unrestricted case:

$$\Delta = (\rho - \delta)[\hat{c} - c] \tag{8}$$

$$\rightarrow \hat{c} - c < 0 \tag{9}$$

$\Delta < 0$ when $\rho, \delta > 0$ (and I showed $\hat{c} < c$), implying that adding innovative-ness to the model actually lowers the expected innovative-ness of the products procured.

This is a key result. When the model allows G and A to value innovation, then the threshold by which A's beliefs must be shifted for it to procure lowers. G can then design a less informative signal in order to induce procurement (i.e., lower β^* means more low fitness products get a positive signal, m_g), diluting the total information available to A and reducing μ_g, and ultimately v_g. In fact, given this, the equilibrium falls short of the socially optimal innovation levels. Let $\tau_H = 1$ be the socially optimal levels of innovation (such that in an ideal world only highly innovative products get procured). If the $E[\tau|\text{procure}, \text{eq}] = \delta + (\rho - \delta)\hat{c}$ then the difference in the expected value of innovation for products procured in equilibrium is always less than the socially optimal levels so long as $\hat{c} < \frac{(\tau_H - \delta)}{\rho - \delta}$ since $\tau_H = 1$ and $\rho, \delta > 0$. This is the case when $\hat{c} < 1$ as I showed previously.

Critically, what allowing G to signal about innovation does in the model is two fold. First, it reduces the threshold by which A's beliefs must be shifted, allowing the government to design less informative signals than in the baseline. Second, it lowers the expected innovative-ness of products that do ultimately get procured, below the socially optimal level (if we assume that it is socially optimal for only highly innovative products to get procured) and below the level one would observe under the baseline model where governments could only signal about fitness.

Lastly, the equilibrium is sensitive to changes in ρ. G's objective function does not factor in ρ unless it changes A's procurement threshold. Since A's procurement threshold is $\hat{c} = \frac{c-\lambda\delta}{1+\lambda(\rho-\delta)}$, then an increase in ρ results in a decrease in $\hat{c}$. This implies that as the correlation between high innovative-ness and high fitness increases, it becomes easier for the government to persuade the agency to procure on a positive message.

5 Conclusion

This paper has attempted to address the question of how and under what circumstances do governments alter the distribution of products and their innovative-ness in the business-to-government market through the use of a Bayesian persuasion model. The goal of model was to illustrate how governments can act as strategically signalers of product level information in order to influence the decisions of actors in the B2G marketplace. The results both the restricted/baseline, alternative, and unrestricted model suggest that when governments care about influencing procurement of the products they believe are the best likely to be successful, they can successfully persuade. However, even with attempts to encourage procurement of only the most likely successful products results in false-positive procurement. Yet, what comparison between the Bayesian Persuasion and PBE analysis reveals, is that pre-commitment of signals by the government is necessary to induce firm entry and to that degree, firm innovation since without entry products are never able to procured at all. This result has important implications for debates about the role of industrial policy as a whole—it suggests that industrial policy, which is a close equivalent to government committing to signals through regulation, is necessary for innovation in the context of the business-to-government marketplace.

Perhaps more importantly, equilibrium analysis reveals that governments do best by limiting what they signal on. While governments can likely forecast the products that will be the most successful (or simply the ones they want to be the most successful), when extra predictions about product-level innovative-ness—that is it's correlation to product success—are added into the signal the expected value of product innovative-ness fails below the socially optimum level. This suggests that if the goal of industrial policies directed at improving product level innovation is to increase procurement (i.e., success) of the most innovative products, then governments do worse by making predictions about product innovative-ness.

This has particular implications for the design of industrial policy as well, particularly in the comparative case between different policy designs. As mentioned previously, the Taiwanese government had success with its large equity stake early in the development of TSMC's semiconductor chips. Other instances of industrial policy in the US and others have not been as successful. Perhaps the difference in innovative outcomes is due not to fundamental research and development differences but in how and what governments signal to markets. Further research should consider this model as a theoretical road map to investigate divergent outcomes in industrial policy. Particularly I think research would benefit from altering the type of model from Bayesian persuasion to one in which governments do not have to pre-commit to signals—hopefully demonstrating how under circumstances where governments might want to obfuscate their true intentions to markets, alternative equilibrium results can occur. I think it'd be particularly interesting too to add dynamic and temporal features to the model to see how industrial policy can be influenced by markets, and not just vice versa.

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A Appendix 1: Proof of Lemma 1

Proof. Consider when $\pi > c$. A's best response to procure, since the costs are lower than their prior beliefs that $\theta = H$. G, knowing this, can commit to sending any signal, since A's priors already favor procuring. G can fully reveal the entire truth, $\alpha^* = \beta^* = 1$, or babble and reveal nothing. Procurement of high fitness products occurs regardless. Optimally, G babbles (non-informative signal), leaving A's posteriors the same as it's priors and maximizing $E[U_G]$ since high fitness product's procurement is secured. While, adding additional information to the signal, does not hurt the government, I assume it prefers not to when it does not have to.

Consider when $\pi < c$. A's best response is not to procure under it's priors. G must design to increase A's posterior belief that $\theta = H$ given $m = g$, and vice versa for $m = b$. G does so by finding α^* that pushes $\mu_g \geq c$ while also setting β^* to $max(Pr(m = g|H))$. The same is the case in the unrestricted model except c is replaced with a different calculated threshold, $\hat{c}$. □

B Appendix 2: Proof of Proposition 1

Proof. Consider that when $\pi < c$, by Lemma 1 G must send a signal that raises A's posteriors to $\mu_g = c$. Given Bayes Rule the following must be true:

$$\frac{\alpha\pi}{\alpha\pi + (1-\beta)(1-\pi)} = c \tag{10}$$

$$\Rightarrow 1 - \beta = \frac{\alpha\pi(1-c)}{c(1-\pi)} \tag{11}$$

G wants to maximize $Pr(m = g|\theta = H) = \alpha$ subject to the constraint on β above and $\beta \in [0, 1]$. Recall G's payoff when A procures on $m = g$ is $E[U_G] = Pr(m = g)|\theta = H) \times Pr(\theta = H) = \alpha\pi$ which is strictly increasing in α. Thus, G prefers to set α as large as possible subject to $\alpha \in [0, 1]$ as well. Then consider when $\alpha^* = 1$, such that solving for β in

Equation 8 gives us:

$$1 - \beta = \frac{\pi(1-c)}{c(1-\pi)} \tag{12}$$

$$\Rightarrow \beta^* = 1 - \frac{\pi(1-c)}{c(1-\pi)} \tag{13}$$

β^* is feasible whenever $\beta^* \in [0,1]$, therefore it must be the case that $\frac{\pi(1-c)}{c(1-\pi)} \geq 0$ which requires $\pi, c > 0$ which is true by assumption. It must also be the case that $\frac{\pi(1-c)}{c(1-\pi)} \leq 1$ for the lower bound on β^* to hold. Thus:

$$\frac{\pi(1-c)}{c(1-\pi)} \leq 1 \tag{14}$$

$$\Rightarrow \pi - \pi c \leq c - c\pi \tag{15}$$

$$\Rightarrow \pi \leq c \tag{16}$$

Equation 13 holds by proposition. Therefore, the optimal signal is $(\alpha^*, \beta^*) = (1, 1 - \frac{\pi(1-c)}{c(1-\pi)})$ implying $Pr(m = g|\theta = H) = 1$, $Pr(m = b|\theta = L) = \beta^*$ and $Pr(m = g|\theta = L) = 1 - \beta^*$, yielding maximum utility for G and indifference over procurement for A since $\mu_g = c$. However, I assume by choice that agencies strictly prefer to procure when the signal is positive.

G does not benefit from deviating from $\mu_g = c$ because of the Bayes Plausibility constraint:

$$E[U_G](\mu_g = c) = Pr(m = g|\theta = H)Pr(\theta = H) = \alpha\pi > E[U_G](\mu_g < c) = 0 \tag{17}$$

The inequality holds because the Agency would never procure when it's beliefs are less than the costs and both $\alpha, \pi > 0$. Similarly,

$$E[U_G](\mu_g > c) = Pr(m = g|\theta = H)1[\mu_g \geq c] + Pr(m = b|\theta = H)1[\mu_b \geq c] \tag{18}$$

$$= \alpha \times 1[\mu_g \geq c] + (1 - \alpha)1[\mu_b \geq c] \tag{19}$$

The right side of Equation 16 is satisfied at $\mu_g = c$ and thus any increase in $\mu_g > c$ only increases G's utility if it also increases $\mu_b > c$, but this violates the Bayes Plausibility constraint. Namely if G increases both μ_g, μ_b then $Pr(m = g)\mu_g + Pr(m = b)\mu_b \neq \pi$, thus G gains nothing from deviating to $\mu_g > c$.

Lastly, posterior beliefs are consistent with the agency's priors given α^*, β^*. First both messages are sent with positive probability such that there are no off-equilibrium path beliefs to specify:

$$\begin{aligned} Pr(m = g) &= \alpha\pi + (1 - \beta)(1 - \pi) \\ &= \pi + \frac{\pi(1 - c)}{c(1 - \pi)}(1 - \pi) = \frac{\pi}{c} > 0 \end{aligned} \tag{20}$$

$$\begin{aligned} Pr(m = b) &= (1 - \alpha)\pi + \beta(1 - \pi) \\ &= 0 + \beta^*(1 - \pi) > 0 \end{aligned} \tag{21}$$

Upon observing $m = g$, A now believes:

$$\begin{aligned} \mu_g &= \frac{Pr(m = g|\theta = H)Pr(\theta = H)}{Pr(m = g)} \\ &= \frac{\alpha^*\pi}{\alpha^*\pi + (1 - \beta^*)(1 - \pi)} \\ &= \frac{\pi}{\pi + (1 - \beta^*)(1 - \pi)}, \text{ subbing in for } \beta^* : \\ &= \frac{\pi}{\pi(1 - 1 - \frac{\pi(1-c)}{c(1-\pi)})(1 - \pi)} \\ &= c \end{aligned} \tag{22}$$

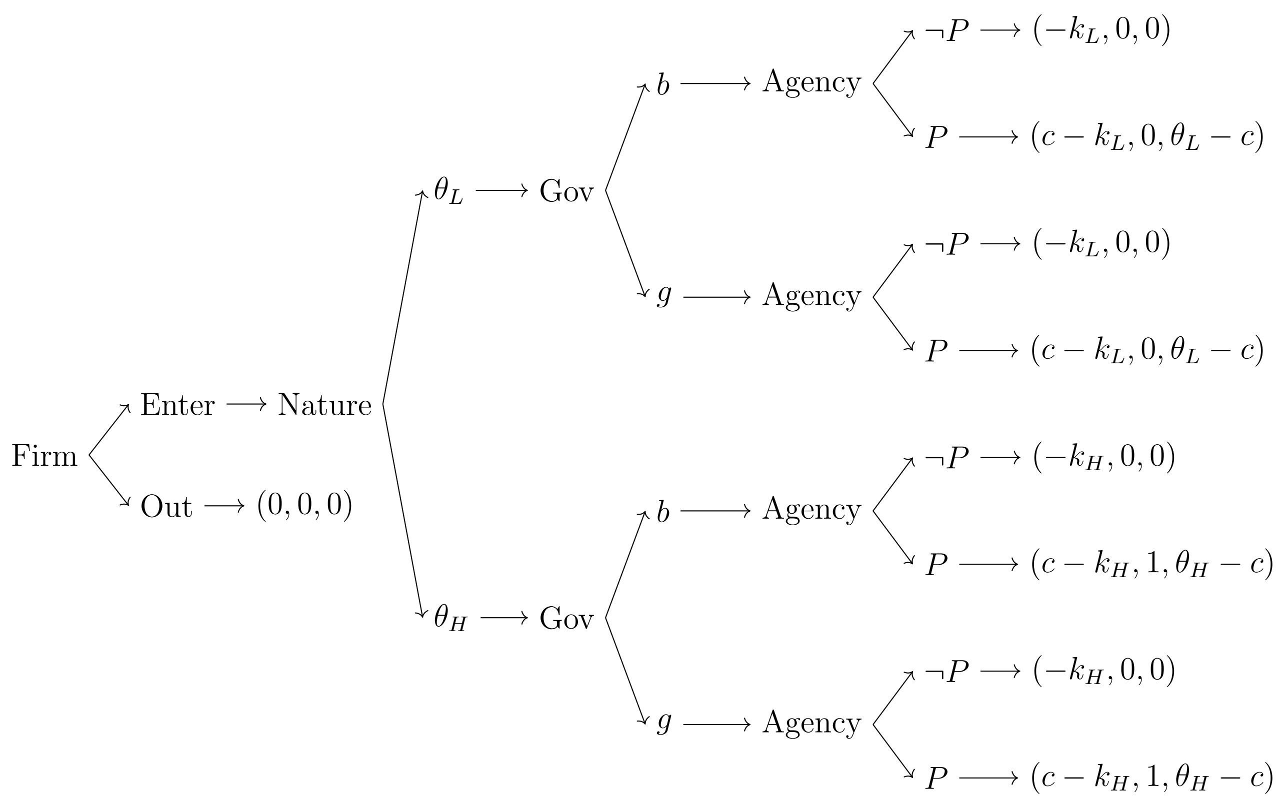
Therefore, A's prior belief yields exactly $\mu_g = c$, as imposed by the equilibrium strategy of

G. Similarly, suppose A observes $m = b$ thus:

$$\begin{aligned} \mu_b &= \frac{Pr(m=b|\theta=H)Pr(\theta=H)}{Pr(m=b)} \\ &= \frac{(1-\alpha^*)\pi}{(1-\alpha^*)\pi+\beta^*(1-\pi)} = 0, \text{ when } \alpha^*=1 \end{aligned} \tag{23}$$

such that A never procures on receiving a negative signal. This is consistent with A's strategy since procuring on a negative signal yields $E[U_A|m = b, \text{procure}] = \mu_b - c = -c$ which is strictly less than the agency's payoff if it did not procure, $E[U_A|b, \text{not procure}] = 0$. □

C Appendix 3: Alternative Model Game Tree/Description



D Appendix 4: Proof of Proposition 2

Proof. Consider A's beliefs about the type of product given signals where product types separate on signals from the government:

$$\mu_A(\theta = H|m = g) = \frac{Pr(m = g|\theta = H)Pr(\theta = H)}{Pr(m = g)} = \frac{\pi}{\pi + 0(1-\pi)} = 1 \tag{24}$$

$$\mu_A(\theta = L|m = b) = \frac{Pr(m = b|\theta = L)Pr(\theta = L)}{Pr(m = L)} = \frac{1-\pi}{1-\pi + 0(\pi)} = 1 \tag{25}$$

Let off-equilibrium path beliefs be that $Pr(\theta = H|m = b) = 0$ and $Pr(\theta = L|m = g) = 0$. For A to want to procure given $m = g$ and not on $m = b$ it must be that:

$$EU_A(\text{procure}|m = g) = \theta_H - c \geq EU_A(\neg\text{procure}|m = g) = 0 \tag{26}$$

$$\Rightarrow \theta_H \geq c \text{ and} \tag{27}$$

$$EU_A(\neg\text{procure}|m = b) = 0 \geq EU_A(\text{procure}|m = b) = \theta_L - c \tag{28}$$

$$\Rightarrow \theta_L \geq c \tag{29}$$

So for sequential rationality to hold for A it must be that $\theta_H > c > \theta_L$. Since I assumed generically that $\theta_H = 1, \theta_L = 0$ and that $c > 0$, then I know this condition holds. The government's choose of $m = g$ for θ_H and $m = b$ for θ_L is obviously sequentially rational because the government only cares about high type products getting procured, and with separating on signals G can ensure only high type products get procured and doesn't gain anything from allowing positive signals about $\theta = L$ products.

Lastly, for the firm's action of entering to be sequentially rational it must be that:

$$EU_F(\neg\text{enter}) = 0 \geq EU_F(\text{enter}) = \pi(c - k_H) + (1-\pi)(-k_L) \tag{30}$$

$$\Rightarrow \pi c \geq \pi k_H + (1-\pi)k_L \tag{31}$$

Consider now the Bayesian Persuasion setting where G pre-commits to a signal that both

A and F observe. Recall from the baseline model from Proposition 1 that $Pr(\text{procure}|\theta = H) = Pr(m = g|\theta = H) = \alpha$ and $Pr(\text{procure}|\theta = L) = Pr(m = g|\theta = L) = 1 - \beta$. Now consider F's payoffs when it considers entering the market:

$$EU_F(\text{enter}) = \pi(c \times Pr(\text{procure}|\theta = H) - k_H) + (1 - \pi)(c \times Pr(\text{procure}|\theta = L) - k_L) \geq \tag{32}$$

$$EU_F(\neg\text{enter}) = 0 \tag{33}$$

Thus F enters the market whenever: $c(\alpha\pi + (1-\pi)(1-\beta)) \geq \pi k_H + (1-\pi)k_L$ where the left hand side of the inequality is the expected contract revenue of a procured product and the right hand side is the expected production costs of a product. G's problem is now to maximize $\alpha, \beta \in [0, 1]$ to ensure that the agency procures on a positive signal and that to ensure the firm enters. This means that G must set $\mu_g > c$ for the agency, while simultaneously ensuring that F's entry condition is met. From the baseline model in Proposition 1, I know that G ensure agency's procurement on positive messages when $\mu_g = c$ and $\alpha^* = 1, \beta^* = 1 - \frac{\pi(1-c)}{c(1-\pi)}$. For α^*, β^* to fulfill the firm's entry condition it must be that:

$$c(\pi\alpha^* + (1 - \pi)(1 - \beta^*) \geq \pi k_H + (1 - \pi)k_L \tag{34}$$

$$\Rightarrow c(\pi + (1 - \pi)(1 - (1 - \frac{\pi(1 - c)}{c(1 - \pi)})) \geq \pi k_H + (1 - \pi)k_L \tag{35}$$

Leaving the right hand side of the inequality alone and simplifying the left hand side gives us the firm's entry condition under Bayesian Persuasion, $\pi \geq \pi k_H + (1 - \pi)k_L$.

Recall the entry condition from the PBE, $\pi c \geq \pi k_H + (1 - \pi)k_L$. With $\pi \in (0, 1)$ and $c \in (0, 1)$, then it must be the case that the left hand side the entry condition for the PBE is smaller than in the Bayesian Persuasion case. $\square$

E Appendix 5: Proof of Proposition 3

Proof. The proof is nearly identical to that in Proposition 1, however, a new procurement threshold, $\hat{c}$ must be derived. Upon observing m, A's posterior beliefs about θ are the same as in Proposition 1:

$$\mu_g = \frac{\alpha\pi}{\alpha\pi + (1-\beta)(1-\pi)} \tag{36}$$

$$\mu_b = \frac{(1-\alpha)\pi}{(1-\alpha)\pi + \beta(1-\pi)} \tag{37}$$

This is because the signal only contains information about θ and τ conditional on θ. Therefore, the agency's beliefs about τ can be conditioned on θ after observing m such that beliefs are a partition of θ, such that now A can update on τ using the law of total probability:

$$Pr(\tau = H|\theta = H, m = g) = Pr(\theta = H|m = g) + Pr(\tau = H|\theta = L, m = g)Pr(\theta = L|m = g) \tag{38}$$

$$Pr(\tau = H|\theta = H, m = b) = Pr(\theta = H|m = b) + Pr(\tau = H|\theta = L, m = b)Pr(\theta = L|m = b) \tag{39}$$

Which, given conditional independence (i.e., that τ doesn't depend on m, given θ) both equations simplify to:

$$Pr(\tau = H|m = g) = \delta + (\rho - \delta)\mu_g = v_g \tag{40}$$

$$Pr(\tau = H|m = b) = \delta + (\rho - \delta)\mu_b = v_b \tag{41}$$

where v_g, v_b are now the posterior probabilities. Since I assumed that $\rho > \delta$ and $\mu_g > \mu_b$ then it must be the case that $v_g > v_b$. Generically, let $v_m = \delta + (\rho - \delta)\mu_m$.

Now consider the agency's payoffs given their beliefs. $E[U_A|m, procure] = \theta + \lambda\tau - c$ and nothing if they don't procure. Thus after observing m, $E[U_A|m, procure) = \mu_m(1) + (1 -$

$\mu_m)0 + \lambda(v_m(1) + (1 - v_m)0) - c$, which reduces to $\mu_m + \lambda(v_m) - c$. Subbing in for v_m gives us: $\mu_m + \lambda(\delta + (\rho - \delta)\mu_m) - c$ which must be greater than 0 for the agency to procure. Thus:

$$\begin{aligned} \mu_m + \lambda(\delta + (\rho - \delta)\mu_m) - c &\geq 0 \\ \mu_m &\geq \frac{c - \delta\lambda}{1 + \lambda(\rho - \delta)} \end{aligned} \tag{42}$$

Thus I define the agency's procurement threshold as $\hat{c} = \frac{c-\delta\lambda}{1+\lambda(\rho-\delta)}$. Now consider the government's payoffs given this and Lemma 1. When A procures after m, $E[U_G|procure, m) = Pr(\theta = H|m) + \Gamma Pr(\tau = H|m)$. Plugging in relevant quantities gives us:

$$\mu_m \times \Gamma v_m \tag{43}$$

$$\Rightarrow \mu_m(1 + \Gamma(\rho - \delta)) + \Gamma\delta \tag{44}$$

$$\text{or when plugging in for A's beliefs} \tag{45}$$

$$(\alpha\pi \times 1[\mu_g \geq \hat{c}] + (1 - \alpha)\pi \times 1[\mu_b \geq \hat{c}] \times (1 + \Gamma(\rho - \delta)) + \Gamma\delta) \times Pr(procure|m) \tag{46}$$

Given Proposition 1, it must be the case that any increase in μ_g must also result in a decrease in μ_b in order for the Bayes Plausibility constraint to hold. Therefore, G's optimal signal raises $\mu_g = \hat{c}$, implying that $\mu_g = \hat{c} > \mu_b$. Thus G seeks to maximize $E[U_G] = \alpha\pi(1 + \Gamma(\rho - \delta)) + \Gamma\delta \times Pr(m = g)$. Like in Proposition 1, G's utility function is increasing in α, therefore G sets α maximally subject to $\alpha \in [0, 1]$, such that $\alpha^* = 1$. Plugging in $\alpha^* = 1$ into the equilibrium condition that $\mu_g = \hat{c}$:

$$\mu_g = \frac{\alpha\pi}{\alpha\pi + (1 - \beta)(1 - \pi)} = \hat{c} \tag{47}$$

$$\hat{c} = \frac{\pi}{\pi + (1 - \beta)(1 - \pi)} \tag{48}$$

$$\beta^* = 1 - \frac{\pi(1 - \hat{c})}{\hat{c}(1 - \pi)} \tag{49}$$

This must hold for $\beta \in [0, 1]$, which is the case whenever $\pi \leq \hat{c}$ and when $\pi \neq 1$ and $\hat{c} \neq 0$.

$\pi \in (0, 1)$ and $\hat{c}$ is non-zero since it is a posterior probability. Lastly, $\pi \leq \hat{c}$ is true by assumption of the Proposition. Beliefs are consistent with best responses because of the Bayes Plausibility constraint (see proof of Proposition 1 for more detail). □

F Appendix 4: AI Declaration

During the preparation of this material, the author used Claude AI for the following reasons: model verification (i.e., math checking), doubling check formulation of more complex ideas (i.e., verifying translation of ideas into formal concepts), equilibrium analysis prediction (i.e., which equilibrium concepts might be more appropriate), grammar and spell check. All inherent modeling choices, assumptions, and interpretations are my own. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the paper.